

Week 5

Raghav Kuinkel (NP069732)

Question no: 1 To display all employee records

Query: CREATE PROCEDURE GetAllEmployees

AS

BEGIN

SELECT * FROM Employee;

END;

EXEC GetAllEmployees;

Output:

Results

Messages

	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City
1	101	Ram Sharma	IT	Developer	65000.00	2022-01-15	Kathmandu
2	102	Sita Karki	HR	Manager	75000.00	2020-05-10	Pokhara
3	103	Hari Thapa	IT	Developer	60000.00	2021-03-12	Kathmandu
4	104	Gita Rai	Finance	Accountant	55000.00	2023-02-18	Biratnagar
5	105	Bikash Gurung	IT	Manager	90000.00	2019-07-25	Pokhara
6	106	Nabin Shrestha	HR	Officer	45000.00	2022-09-15	Kathmandu
7	107	Anita Lama	Finance	Manager	85000.00	2018-11-05	Lalitpur
8	108	Roshan KC	Marketing	Executive	50000.00	2021-12-01	Pokhara
9	109	Puja Basnet	Marketing	Manager	80000.00	2020-08-20	Kathmandu
10	110	Kiran Adhikari	IT	Officer	48000.00	2023-04-10	Biratnagar

Question no 2 To display the record of employee accepting the employee ID

Query:

CREATE PROCEDURE GetEmployeeByID

@EmpID INT

AS

BEGIN

SELECT * FROM Employee

WHERE EmployeeID = @EmpID;

END;

EXEC GetEmployeeByID @EmpID = 103;

Results

Messages

	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City
1	103	Hari Thapa	IT	Developer	60000.00	2021-03-12	Kathmandu

Question no: 3 That accepts a salary amount and displays employees earning more than that amount

Query:

```

CREATE PROCEDURE GetEmployeesAboveSalary
    @MinSalary DECIMAL(10,2)
AS
BEGIN
    SELECT * FROM Employee
    WHERE Salary > @MinSalary;
END;
EXEC GetEmployeesAboveSalary @MinSalary = 60000;

```

Results		Messages						
	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City	
1	101	Ram Sharma	IT	Developer	65000.00	2022-01-15	Kathmandu	
2	102	Sita Karki	HR	Manager	75000.00	2020-05-10	Pokhara	
3	105	Bikash Gurung	IT	Manager	90000.00	2019-07-25	Pokhara	
4	107	Anita Lama	Finance	Manager	85000.00	2018-11-05	Lalitpur	
5	109	Puja Basnet	Marketing	Manager	80000.00	2020-08-20	Kathmandu	

Question no: 4 To display total number of employees

Query:

```

CREATE PROCEDURE GetTotalEmployees
AS
BEGIN
    SELECT COUNT(*) AS TotalEmployees FROM Employee;
END;
EXEC GetTotalEmployees;

```

Results		Messages						
	TotalEmployees							
1	10							

Question no: 5 That display employee ordered by department and salary in descending order

Query:

```

CREATE PROCEDURE GetEmployeesSorted
AS
BEGIN
    SELECT * FROM Employee
    ORDER BY Department ASC, Salary DESC;
END;
EXEC GetEmployeesSorted;

```

Results Messages							
	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City
1	107	Anita Lama	Finance	Manager	85000.00	2018-11-05	Lalitpur
2	104	Gita Rai	Finance	Accountant	55000.00	2023-02-18	Biratnagar
3	102	Sita Karki	HR	Manager	75000.00	2020-05-10	Pokhara
4	106	Nabin Shrestha	HR	Officer	45000.00	2022-09-15	Kathmandu
5	105	Bikash Gurung	IT	Manager	90000.00	2019-07-25	Pokhara
6	101	Ram Sharma	IT	Developer	65000.00	2022-01-15	Kathmandu
7	103	Hari Thapa	IT	Developer	60000.00	2021-03-12	Kathmandu
8	110	Kiran Adhikari	IT	Officer	48000.00	2023-04-10	Biratnagar

Question no 6 To display top 3 highest paid employees

Query:

```
CREATE PROCEDURE GetTop3HighestPaid
```

```
AS
```

```
BEGIN
```

```
    SELECT TOP 3 * FROM Employee
```

```
    ORDER BY Salary DESC;
```

```
END;
```

```
EXEC GetTop3HighestPaid;
```

Results Messages							
	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City
1	105	Bikash Gurung	IT	Manager	90000.00	2019-07-25	Pokhara
2	107	Anita Lama	Finance	Manager	85000.00	2018-11-05	Lalitpur
3	109	Puja Basnet	Marketing	Manager	80000.00	2020-08-20	Kathmandu

Question no: 7 To display employee whose name contains 'sh'

Query:

```
CREATE PROCEDURE GetEmployeesByNamePattern
AS
BEGIN
    SELECT * FROM Employee
    WHERE FullName LIKE '%sh%';
END;
EXEC GetEmployeesByNamePattern;
```

Results

Messages

	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City
1	101	Ram Sharma	IT	Developer	65000.00	2022-01-15	Kathmandu
2	105	Bikash Gurung	IT	Manager	90000.00	2019-07-25	Pokhara
3	106	Nabin Shrestha	HR	Officer	45000.00	2022-09-15	Kathmandu
4	108	Roshan KC	Marketing	Executive	50000.00	2021-12-01	Pokhara

Question no: 8 To display employee from IT and Finance Department

Query:

```
CREATE PROCEDURE GetITAndFinanceEmployees
AS
BEGIN
    SELECT * FROM Employee
    WHERE Department IN ('IT', 'Finance');
END;
EXEC GetITAndFinanceEmployees;
```

Results

Messages

	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City
1	101	Ram Sharma	IT	Developer	65000.00	2022-01-15	Kathmandu
2	103	Hari Thapa	IT	Developer	60000.00	2021-03-12	Kathmandu
3	104	Gita Rai	Finance	Accountant	55000.00	2023-02-18	Biratnagar
4	105	Bikash Gurung	IT	Manager	90000.00	2019-07-25	Pokhara
5	107	Anita Lama	Finance	Manager	85000.00	2018-11-05	Lalitpur
6	110	Kiran Adhikari	IT	Officer	48000.00	2023-04-10	Biratnagar

Question no 9 To display employee whose designation is not Manager

Query:

```
CREATE PROCEDURE GetNonManagers
AS
BEGIN
    SELECT * FROM Employee
    WHERE Designation <> 'Manager';
END;
EXEC GetNonManagers;
```

Results		Messages					
	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City
1	101	Ram Sharma	IT	Developer	65000.00	2022-01-15	Kathmandu
2	103	Hari Thapa	IT	Developer	60000.00	2021-03-12	Kathmandu
3	104	Gita Rai	Finance	Accountant	55000.00	2023-02-18	Biratnagar
4	106	Nabin Shrestha	HR	Officer	45000.00	2022-09-15	Kathmandu
5	108	Roshan KC	Marketing	Executive	50000.00	2021-12-01	Pokhara
6	110	Kiran Adhikari	IT	Officer	48000.00	2023-04-10	Biratnagar

Question no: 10 To display minimum and maximum salary

Query:

```
CREATE PROCEDURE GetMinMaxSalary
AS
BEGIN
    SELECT
        MIN(Salary) AS MinimumSalary,
        MAX(Salary) AS MaximumSalary
    FROM Employee;
END;

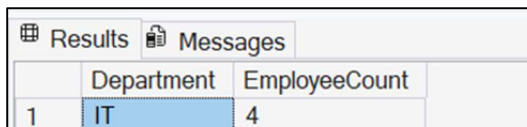
EXEC GetMinMaxSalary;
```

Results		Messages	
	MinimumSalary	MaximumSalary	
1	45000.00	90000.00	

Question no: 11 To display departments having more than 2 employees

Query:

```
CREATE PROCEDURE GetLargeDepartments
AS
BEGIN
    SELECT Department, COUNT(*) AS EmployeeCount
    FROM Employee
    GROUP BY Department
    HAVING COUNT(*) > 2;
END;
EXEC GetLargeDepartments;
```

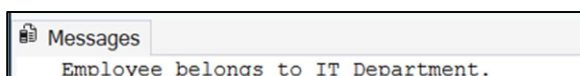


	Department	EmployeeCount
1	IT	4

Question no: 12 To display whether an employee belongs to IT department

Query:

```
CREATE PROCEDURE CheckITDepartment
    @EmpID INT
AS
BEGIN
    DECLARE @Dept VARCHAR(30);
    SELECT @Dept = Department
    FROM Employee
    WHERE EmployeeID = @EmpID;
    IF @Dept = 'IT'
        PRINT 'Employee belongs to IT Department.';
    ELSE IF @Dept IS NULL
        PRINT 'Employee not found.';
    ELSE
        PRINT 'Employee does not belong to IT Department. Department: ' + @Dept;
END;
EXEC CheckITDepartment @EmpID = 101;
```



Messages
Employee belongs to IT Department.

Question no 13 To insert an employee if EmployeeID does not exist and display all employees

Query:

```
CREATE PROCEDURE InsertEmployeeIfNotExists
```

```
    @EmpID    INT,
```

```
    @FullName  VARCHAR(50),
```

```
    @Department VARCHAR(30),
```

```
    @Designation VARCHAR(30),
```

```
    @Salary    DECIMAL(10,2),
```

```
    @JoiningDate DATE,
```

```
    @City      VARCHAR(30)
```

```
AS
```

```
BEGIN
```

```
    IF NOT EXISTS (SELECT 1 FROM Employee WHERE EmployeeID = @EmpID)
```

```
    BEGIN
```

```
        INSERT INTO Employee
```

```
        VALUES (@EmpID, @FullName, @Department, @Designation, @Salary, @JoiningDate,
```

```
@City);
```

```
        PRINT 'Employee inserted successfully';
```

```
    END
```

```
ELSE
```

```
    PRINT 'EmployeeID already exists. No record inserted.';
```

```
    SELECT * FROM Employee;
```

```
END;
```

```
EXEC InsertEmployeeIfNotExists
```

```
    @EmpID=111, @FullName='Sunita Tamang', @Department='HR',
```

```
    @Designation='Officer', @Salary=47000,
```

```
    @JoiningDate='2024-01-10', @City='Kathmandu';
```

Results		Messages					
	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City
1	101	Ram Sharma	IT	Developer	65000.00	2022-01-15	Kathmandu
2	102	Sita Karki	HR	Manager	75000.00	2020-05-10	Pokhara
3	103	Hari Thapa	IT	Developer	60000.00	2021-03-12	Kathmandu
4	104	Gita Rai	Finance	Accountant	55000.00	2023-02-18	Biratnagar
5	105	Bikash Gurung	IT	Manager	90000.00	2019-07-25	Pokhara
6	106	Nabin Shrestha	HR	Officer	45000.00	2022-09-15	Kathmandu
7	107	Anita Lama	Finance	Manager	85000.00	2018-11-05	Lalitpur
8	108	Roshan KC	Marketing	Executive	50000.00	2021-12-01	Pokhara

Question no: 14 TO increase the salary of all employees by a given percentage

Query:

```
CREATE PROCEDURE IncreaseSalaryByPercentage
```

```
    @Percentage DECIMAL(5,2)
```

```
AS
```

```
BEGIN
```

```
    UPDATE Employee
```

```
    SET Salary = Salary + (Salary * @Percentage / 100);
```

```
    PRINT 'Salaries updated by ' + CAST(@Percentage AS VARCHAR) + '%.';
```

```
    SELECT EmployeeID, FullName, Salary FROM Employee;
```

```
END;
```

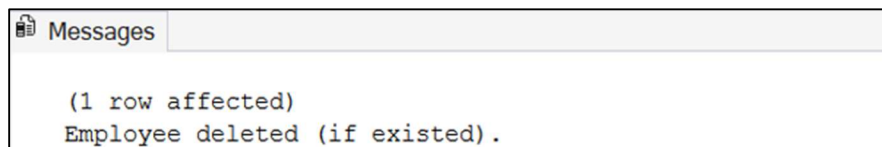
```
EXEC IncreaseSalaryByPercentage @Percentage = 10;
```

Results		Messages	
	EmployeeID	FullName	Salary
1	101	Ram Sharma	71500.00
2	102	Sita Karki	82500.00
3	103	Hari Thapa	66000.00
4	104	Gita Rai	60500.00
5	105	Bikash Gurung	99000.00
6	106	Nabin Shrestha	49500.00
7	107	Anita Lama	93500.00
8	108	Roshan KC	55000.00

Question no: 15 To delete employee by EmployeeID 16. That checks existence before deleting employee

Query:

```
CREATE PROCEDURE DeleteEmployee
    @EmpID INT
AS
BEGIN
    DELETE FROM Employee
    WHERE EmployeeID = @EmpID;
    PRINT 'Employee deleted (if existed).';
END;
EXEC DeleteEmployee @EmpID = 110;
```



Question no: 16 Check existence before deleting employee

Query:

```
CREATE PROCEDURE DeleteEmployeeIfExists
    @EmpID INT
AS
BEGIN
    IF EXISTS (SELECT 1 FROM Employee WHERE EmployeeID = @EmpID)
    BEGIN
        DELETE FROM Employee WHERE EmployeeID = @EmpID;
        PRINT 'Employee with ID ' + CAST(@EmpID AS VARCHAR) + ' has been deleted.';
    END
    ELSE
        PRINT 'Employee with ID ' + CAST(@EmpID AS VARCHAR) + ' does not exist.';
END;

EXEC DeleteEmployeeIfExists @EmpID = 110;
```



Scenario Based:

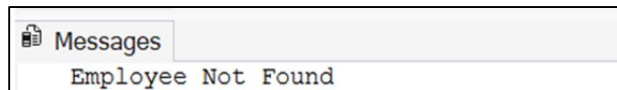
1. Create a stored procedure that

- Accepts EmployeeID as input
- Display Employee Details if found
- Display “Employee Not Found” if the employee does not exist

Query:

```
CREATE PROCEDURE SearchEmployee
    @EmpID INT
AS
BEGIN
    IF EXISTS (SELECT 1 FROM Employee WHERE EmployeeID = @EmpID)
        SELECT * FROM Employee WHERE EmployeeID = @EmpID;
    ELSE
        PRINT 'Employee Not Found';
END;

EXEC SearchEmployee @EmpID = 999;
EXEC SearchEmployee @EmpID = 102;
```



2. New Employee Recruitment

- Accepts employee information
- Checks whether EmployeeID already exists
- Insert the employee if EmployeeID is unique
- Otherwise display an error message

Query:

```
CREATE PROCEDURE RecruitEmployee
    @EmpID INT,
    @FullName VARCHAR(50),
    @Department VARCHAR(30),
    @Designation VARCHAR(30),
    @Salary DECIMAL(10,2),
    @JoiningDate DATE,
```

```

    @City    VARCHAR(30)
AS
BEGIN
    IF EXISTS (SELECT 1 FROM Employee WHERE EmployeeID = @EmpID)
    BEGIN
        PRINT 'Error: EmployeeID ' + CAST(@EmpID AS VARCHAR) + ' already exists.
Cannot insert duplicate.';
    END
    ELSE
    BEGIN
        INSERT INTO Employee
        VALUES (@EmpID, @FullName, @Department, @Designation, @Salary,
@JoiningDate, @City);
        PRINT 'New employee recruited successfully!';
        SELECT * FROM Employee WHERE EmployeeID = @EmpID;
    END
END;

EXEC RecruitEmployee
    @EmpID=112, @FullName='Anil Bhandari', @Department='IT',
    @Designation='Developer', @Salary=62000,
    @JoiningDate='2024-05-01', @City='Pokhara';

-- Try inserting a duplicate:
EXEC RecruitEmployee
    @EmpID=101, @FullName='Duplicate Test', @Department='IT',
    @Designation='Developer', @Salary=50000,
    @JoiningDate='2024-05-01', @City='Kathmandu';

```

Results Messages							
	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City
1	112	Anil Bhandari	IT	Developer	62000.00	2024-05-01	Pokhara

Try inserting a duplicate.

Messages	
Error:	EmployeeID 101 already exists. Cannot insert duplicate.

3. Employee Bonus System

- **Accepts EmployeeID**
- **Calculate bonus according to designation**
- **Display Employee Name and Bonus amount**

Query:

```
DROP PROCEDURE IF EXISTS sp_EmployeeBonus;  
GO
```

```
CREATE PROCEDURE sp_EmployeeBonus  
    @EmployeeID INT  
AS  
BEGIN  
    DECLARE @Salary DECIMAL(10,2);  
    DECLARE @Designation VARCHAR(30);  
    DECLARE @Bonus DECIMAL(10,2);
```

```
    IF EXISTS  
    (  
        SELECT *  
        FROM Employee  
        WHERE EmployeeID = @EmployeeID  
    )
```

```
    BEGIN  
        SELECT  
            @Salary = Salary,  
            @Designation = Designation  
        FROM Employee  
        WHERE EmployeeID = @EmployeeID;
```

```
        IF @Designation = 'Manager'  
            SET @Bonus = @Salary * 0.20;  
        ELSE IF @Designation = 'Developer'  
            SET @Bonus = @Salary * 0.15;  
        ELSE  
            SET @Bonus = @Salary * 0.10;
```

```
    SELECT  
        EmployeeID,
```

```

        FullName,
        Designation,
        Salary,
        @Bonus AS BonusAmount
    FROM Employee
    WHERE EmployeeID = @EmployeeID;
END
ELSE
BEGIN
    SELECT 'Employee Not Found' AS Result;
END
END;
GO

```

```
EXEC sp_EmployeeBonus 105;
```

Results		Messages			
	EmployeeID	FullName	Designation	Salary	BonusAmount
1	105	Bikash Gurung	Manager	1900239.95	380047.99

4. Department Salary Report

- Display department wise total salary
- Display department wise average salary

Query:

```

Display Department-wise Total Salary and Average Salary
DROP PROCEDURE IF EXISTS sp_DepartmentSalaryReport;
GO

```

```

CREATE PROCEDURE sp_DepartmentSalaryReport
AS
BEGIN
    SELECT
        Department,
        SUM(Salary) AS TotalSalary,
        AVG(Salary) AS AverageSalary
    FROM Employee
    GROUP BY Department;

    SELECT * FROM Employee;

```

```

END;
GO
EXEC sp_DepartmentSalaryReport;

```

Results		Messages	
	Department	TotalSalary	AverageSalary
1	Finance	2955928.99	1477964.495000
2	HR	2533653.66	1266826.830000
3	IT	4383324.47	1095831.117500
4	Marketing	2744791.12	1372395.560000

	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City
1	101	Ramesh Nepal	IT	Officer	55000.00	2024-01-01	Kathmandu
2	102	Sita Karki	HR	Manager	1583533.50	2020-05-10	Pokhara
3	103	Hari Thapa	IT	Developer	1266826.67	2021-03-12	Kathmandu
4	104	Gita Rai	Finance	Accountant	1161257.85	2023-02-18	Biratnagar
5	105	Bikash Gurung	IT	Manager	1900239.95	2019-07-25	Pokhara
6	106	Nabin Shrestha	HR	Officer	950120.16	2022-09-15	Kathmandu
7	107	Anita Lama	Finance	Manager	1794671.14	2018-11-05	Lalitpur
8	108	Roshan KC	Marketing	Executive	1055688.95	2021-12-01	Pokhara
9	109	Puja Basnet	Marketing	Manager	1689102.17	2020-08-20	Kathmandu
10	111	Ramesh Nepal	IT	Officer	1161257.85	2024-01-01	Kathmandu

5. Annual Salary Calculator

- Use a WHILE loop.
- Calculates annual salary (Salary × 12) for each employee.
- Displays EmployeeID, Name and Annual Salary

Query:

```

DROP PROCEDURE IF EXISTS sp_AnnualSalary;
GO

```

```

CREATE PROCEDURE sp_AnnualSalary
AS
BEGIN
    DECLARE @ID INT;

    CREATE TABLE #AnnualSalaryReport
    (
        EmployeeID INT,
        FullName VARCHAR(50),
        AnnualSalary DECIMAL(12,2)
    );

```

```

SET @ID =
(
    SELECT MIN(EmployeeID)
    FROM Employee
);

WHILE @ID IS NOT NULL
BEGIN
    INSERT INTO #AnnualSalaryReport
    SELECT
        EmployeeID,
        FullName,
        Salary * 12
    FROM Employee
    WHERE EmployeeID = @ID;

    SET @ID =
    (
        SELECT MIN(EmployeeID)
        FROM Employee
        WHERE EmployeeID > @ID
    );
END

SELECT * FROM #AnnualSalaryReport;

SELECT * FROM Employee;

DROP TABLE #AnnualSalaryReport;
END;
GO

EXEC sp_AnnualSalary;

```


 Results Messages

	EmployeeID	FullName	AnnualSalary				
1	101	Ramesh Nepal	660000.00				
2	102	Sita Karki	19002402.00				
3	103	Hari Thapa	15201920.04				
4	104	Gita Rai	13935094.20				
5	105	Bikash Gurung	22802879.40				
6	106	Nabin Shrestha	11401441.92				
7	107	Anita Lama	21536053.68				
8	108	Roshan KC	12668267.40				
9	109	Puja Basnet	20260226.04				

	EmployeeID	FullName	Department	Designation	Salary	JoiningDate	City
1	101	Ramesh Nepal	IT	Officer	55000.00	2024-01-01	Kathmandu
2	102	Sita Karki	HR	Manager	1583533.50	2020-05-10	Pokhara
3	103	Hari Thapa	IT	Developer	1266826.67	2021-03-12	Kathmandu
4	104	Gita Rai	Finance	Accountant	1161257.85	2023-02-18	Biratnagar
5	105	Bikash Gurung	IT	Manager	1900239.95	2019-07-25	Pokhara
6	106	Nabin Shrestha	HR	Officer	950120.16	2022-09-15	Kathmandu
7	107	Anita Lama	Finance	Manager	1794671.14	2018-11-05	Lalitpur
8	108	Roshan KC	Marketing	Executive	1055688.95	2021-12-01	Pokhara
9	109	Puja Basnet	Marketing	Manager	1689102.17	2020-08-20	Kathmandu
10	111	Ramesh Nepal	IT	Officer	1161257.85	2024-01-01	Kathmandu